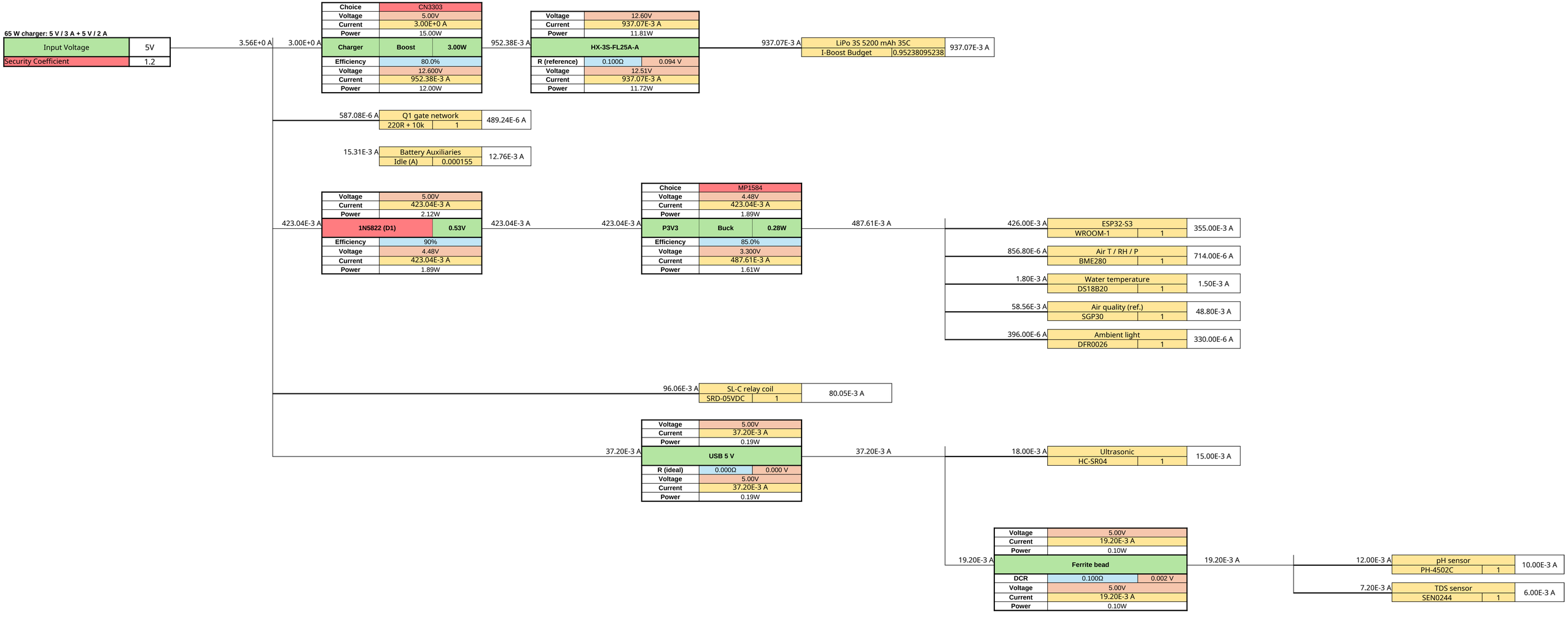
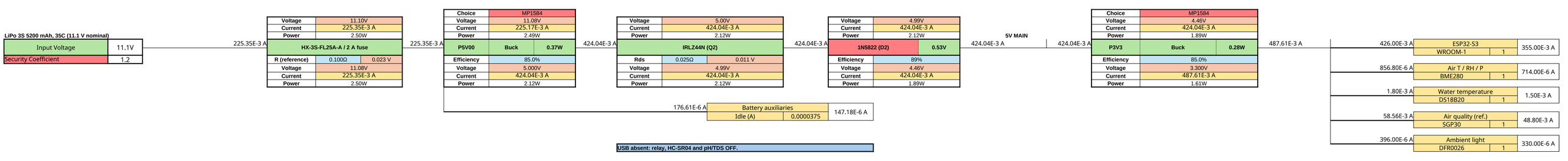

65 W charger: 5 V / 3 A + 5 V / 2 A
Input Voltage 5V
Security Coefficient 1.2
3.56E+0 A
3.00E+0 A
Choice CN3303
Voltage 5.00V
Current 3.00E+0 A
Power 15.00W
Charger Boost 3.00W
Efficiency 80.0%
Voltage 12.600V
Current 952.38E-3 A
Power 12.00W
952.38E-3 A
Voltage 12.60V
Current 937.07E-3 A
Power 11.81W
HX-3S-FL25A-A
R (reference) 0.100Ω 0.094 V
Voltage 12.51V
Current 937.07E-3 A
Power 11.72W
937.07E-3 A
LiPo 3S 5200 mAh 35C
I-Boost Budget 0.95238095238
937.07E-3 A
587.08E-6 A
Q1 gate network
220R + 10k 1
489.24E-6 A
15.31E-3 A
Battery Auxiliaries
Idle (A) 0.000155
12.76E-3 A
Voltage 5.00V
Current 423.04E-3 A
Power 2.12W
423.04E-3 A
1N5822 (D1) 0.53V
Efficiency 90%
Voltage 4.48V
Current 423.04E-3 A
Power 1.89W
423.04E-3 A
Choice MP1584
Voltage 4.48V
Current 423.04E-3 A
Power 1.89W
423.04E-3 A
P3V3 Buck 0.28W
Efficiency 85.0%
Voltage 3.300V
Current 487.61E-3 A
Power 1.61W
487.61E-3 A
426.00E-3 A
ESP32-S3
WROOM-1 1
355.00E-3 A
856.80E-6 A
Air T / RH / P
BME280 1
714.00E-6 A
1.80E-3 A
Water temperature
DS18B20 1
1.50E-3 A
58.56E-3 A
Air quality (ref.)
SGP30 1
48.80E-3 A
396.00E-6 A
Ambient light
DFR0026 1
330.00E-6 A
96.06E-3 A
SL-C relay coil
SRD-05VDC 1
80.05E-3 A
Voltage 5.00V
Current 37.20E-3 A
Power 0.19W
37.20E-3 A
USB 5 V
R (ideal) 0.000Ω 0.000 V
Voltage 5.00V
Current 37.20E-3 A
Power 0.19W
37.20E-3 A
18.00E-3 A
Ultrasonic
HC-SR04 1
15.00E-3 A
Voltage 5.00V
Current 19.20E-3 A
Power 0.10W
19.20E-3 A
Ferrite bead
DCR 0.100Ω 0.002 V
Voltage 5.00V
Current 19.20E-3 A
Power 0.10W
19.20E-3 A
12.00E-3 A
pH sensor
PH-4502C 1
10.00E-3 A
7.20E-3 A
TDS sensor
SEN0244 1
6.00E-3 A


LiPo 3S 5200 mAh, 35C (11.1 V nominal)
Input Voltage
11.1V
Security Coefficient
1.2
225.35E-3 A
Voltage
11.10V
Current
225.35E-3 A
Power
2.50W
HX-3S-FL25A-A / 2 A fuse
R (reference)
0.100Ω
0.023 V
Voltage
11.08V
Current
225.35E-3 A
Power
2.50W
225.35E-3 A
Choice
MP1584
Voltage
11.08V
Current
225.17E-3 A
Power
2.49W
P5V00
Buck
0.37W
Efficiency
85.0%
Voltage
5.000V
Current
424.04E-3 A
Power
2.12W
424.04E-3 A
Voltage
5.00V
Current
424.04E-3 A
Power
2.12W
IRLZ44N (Q2)
Rds
0.025Ω
0.011 V
Voltage
4.99V
Current
424.04E-3 A
Power
2.12W
424.04E-3 A
Voltage
4.99V
Current
424.04E-3 A
Power
2.12W
1N5822 (D2)
0.53V
Efficiency
89%
Voltage
4.46V
Current
424.04E-3 A
Power
1.89W
424.04E-3 A
5V MAIN
424.04E-3 A
Choice
MP1584
Voltage
4.46V
Current
424.04E-3 A
Power
1.89W
P3V3
Buck
0.28W
Efficiency
85.0%
Voltage
3.300V
Current
487.61E-3 A
Power
1.61W
487.61E-3 A
426.00E-3 A
ESP32-S3
WROOM-1
1
355.00E-3 A
856.80E-6 A
Air T / RH / P
BME280
1
714.00E-6 A
1.80E-3 A
Water temperature
DS18B20
1
1.50E-3 A
58.56E-3 A
Air quality (ref.)
SGP30
1
48.80E-3 A
396.00E-6 A
Ambient light
DFR0026
1
330.00E-6 A
176.61E-6 A
Battery auxiliaries
Idle (A)
0.0000375
147.18E-6 A
USB absent: relay, HC-SR04 and pH/TDS OFF.